**Program 6: Continuous Integration with Jenkins:** Setting Up a CI Pipeline**,** Integrating Jenkins with Maven/Gradle**,** Running Automated Builds and Tests

**Objective**

To implement Continuous Integration (CI) using Jenkins by integrating it with Maven/Gradle and executing automated builds and tests for a Java project developed in IntelliJ IDEA.

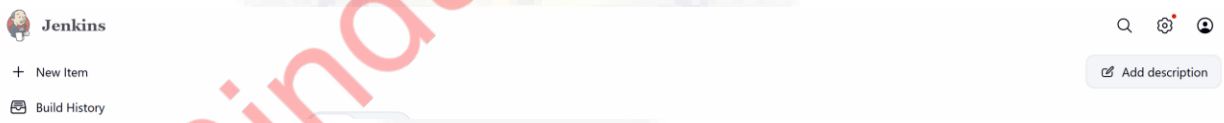
1. Execute the below command to start the Jenkins server

```
D:\>cd Jenkins

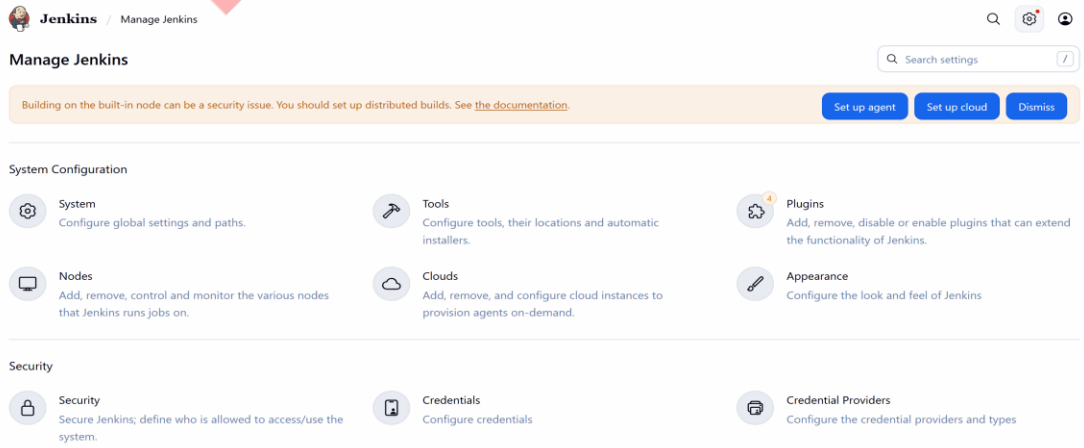
D:\Jenkins>java -jar jenkins.war --httpPort=9090
Running from: D:\Jenkins\jenkins.war
webroot: C:\Users\BINDUSHREE V\.jenkins\war
2026-03-15 15:38:36.817+0000 [id=3]     INFO    winstone.Logger#logInternal: Beginning extraction from war file
2026-03-15 15:38:36.930+0000 [id=3]     WARNING o.e.j.ee9.nested.ContextHandler#setContextPath: Empty contextPath
2026-03-15 15:38:37.061+0000 [id=3]     INFO    org.eclipse.jetty.server.Server#doStart: jetty-12.1.5; built: 2025-12-03
T22:18:24.732Z; git: 4905d09d7b6801e792d7b73946cf7b66e4be25d6; jvm 25.0.2+10-LTS-69
2026-03-15 15:38:38.834+0000 [id=3]     INFO    o.e.j.e.w.StandardDescriptorProcessor#visitServlet: NO JSP Support for /
, did not find org.eclipse.jetty.ee9.jsp.JettyJspServlet
2026-03-15 15:38:39.006+0000 [id=3]     INFO    o.e.j.s.DefaultSessionIdManager#doStart: Session workerName=node0
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::objectFieldOffset has been called by com.thoughtworks.xstream.converters.reflection.SunUnsafeR
eflectionProvider (file:/C:/Users/BINDUSHREE%20V/.jenkins/war/WEB-INF/lib/xstream-1.4.21.jar)
WARNING: Please consider reporting this to the maintainers of class com.thoughtworks.xstream.converters.reflection.SunUn
safeReflectionProvider
WARNING: sun.misc.Unsafe::objectFieldOffset will be removed in a future release
```

**After that** Open browser and go to: http://localhost:9090

2. Configure Maven and Gradle in Jenkins: go to Manage Jenkins



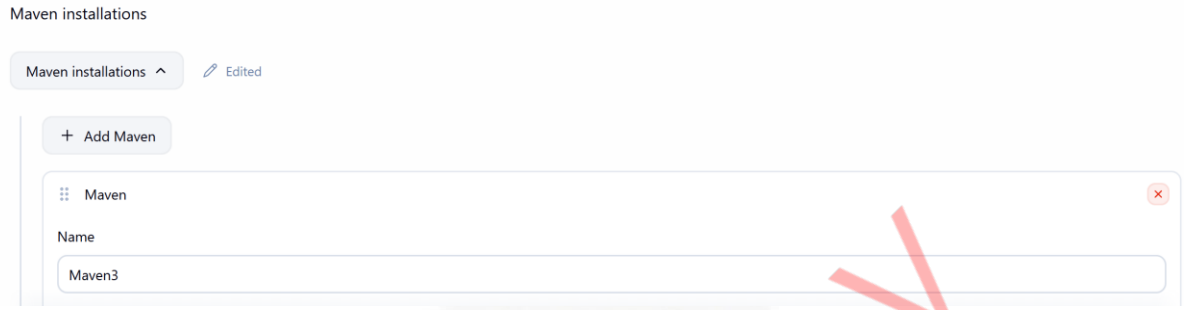

3. Click Global Tool Configuration




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4. Scroll to Maven Installations and gradle Installations:

   Add Maven




Click on save

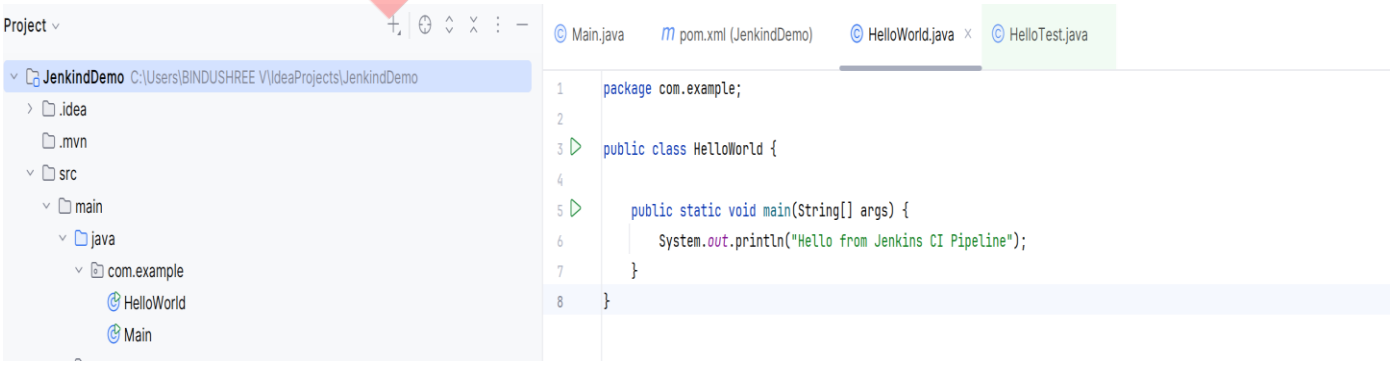
5. Create Java Project in IntelliJ IDEA->New Project→Maven
6. Create Sample Java Program

   HelloWorld.java

```
package com.example;

public class HelloWorld {

    public static void main(String[] args) {
        System.out.println("Hello from Jenkins CI Pipeline");
    }
}
```




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7. Create Test Case

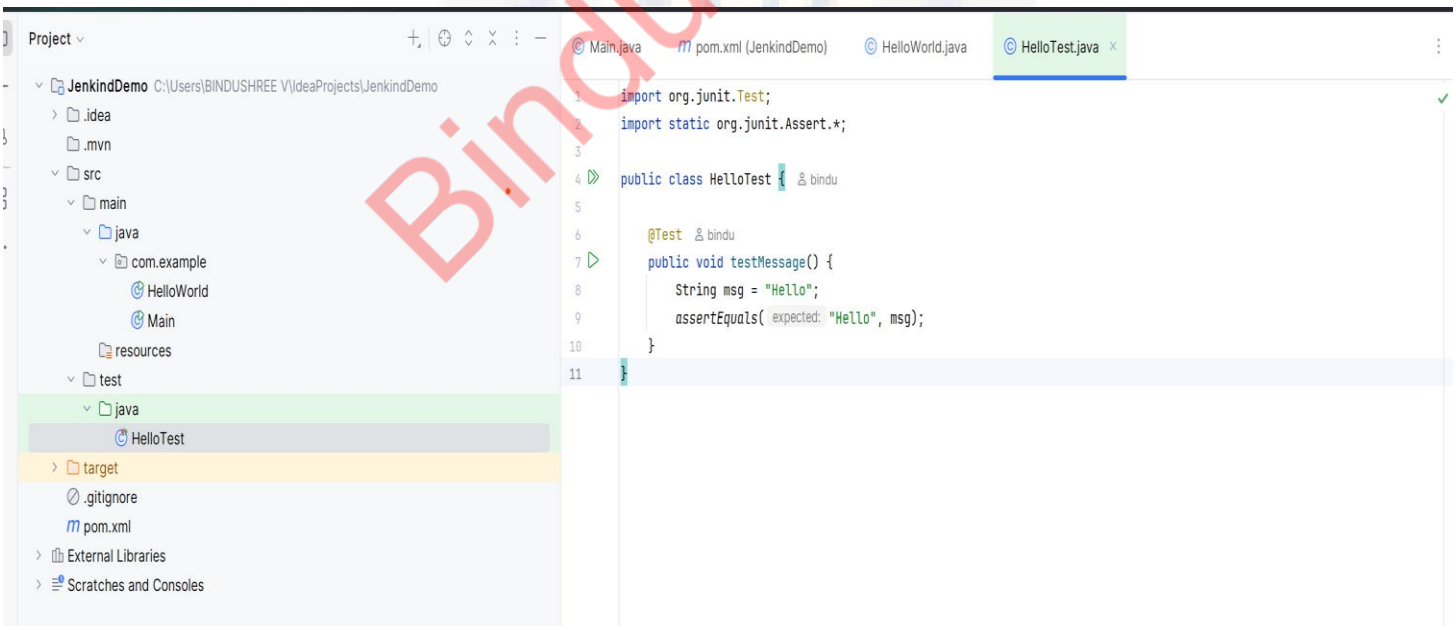
Create a test file

HelloTest.java

```
import org.junit.Test;
import static org.junit.Assert.*;

public class HelloTest {

   @Test
   public void testMessage() {
      String msg = "Hello";
      assertEquals("Hello", msg);
   }
}
```




7. Add the dependency to the pom file and sync the project

<dependencies>

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```
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.13.2</version>
      <scope>test</scope>
    </dependency>

</dependencies>
```

Build project by running the command : **mvn clean install**

8. **Push the files to the github if the maven build is successful**

**git init**

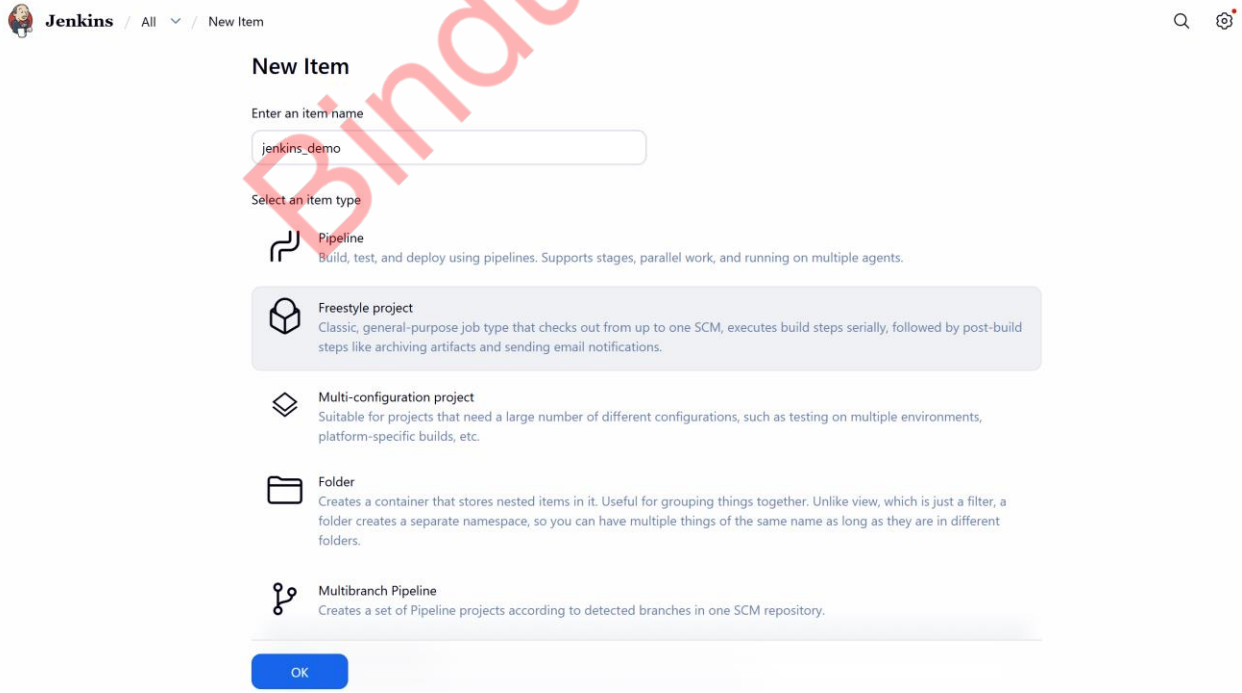
**git add .**

**git commit -m "My Jenkins Build"**

**git remote add origin "add your github URL"**

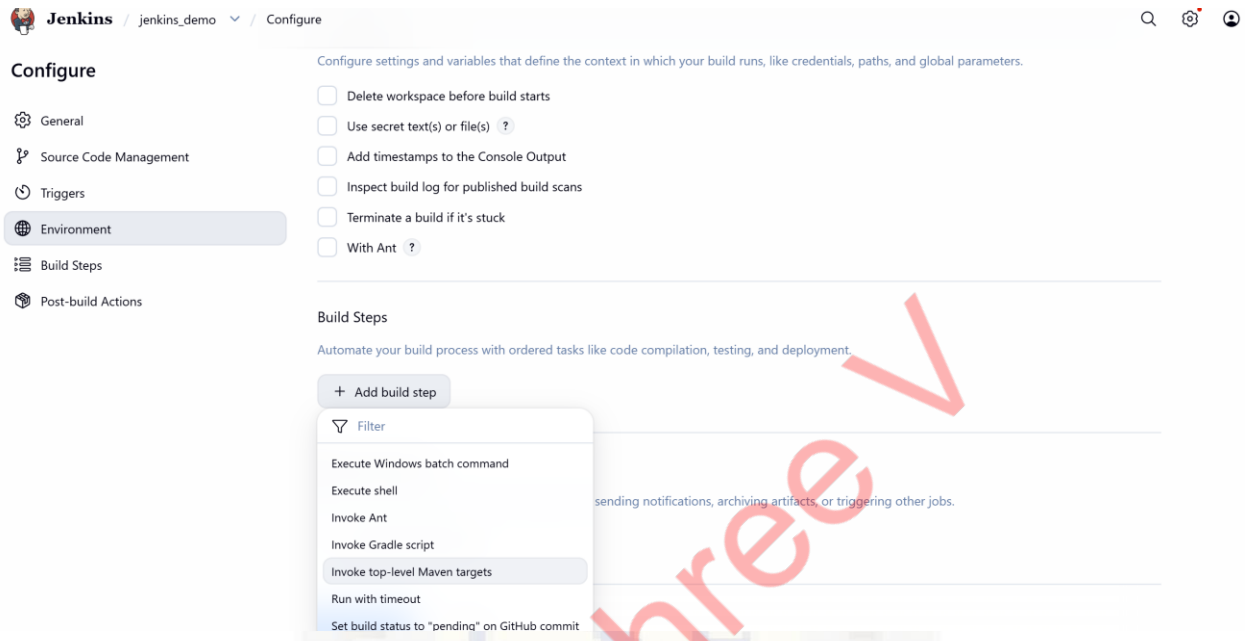
**git push -u origin master**

9. New Item->Enter name->Select Freestyle Project and Click **OK**

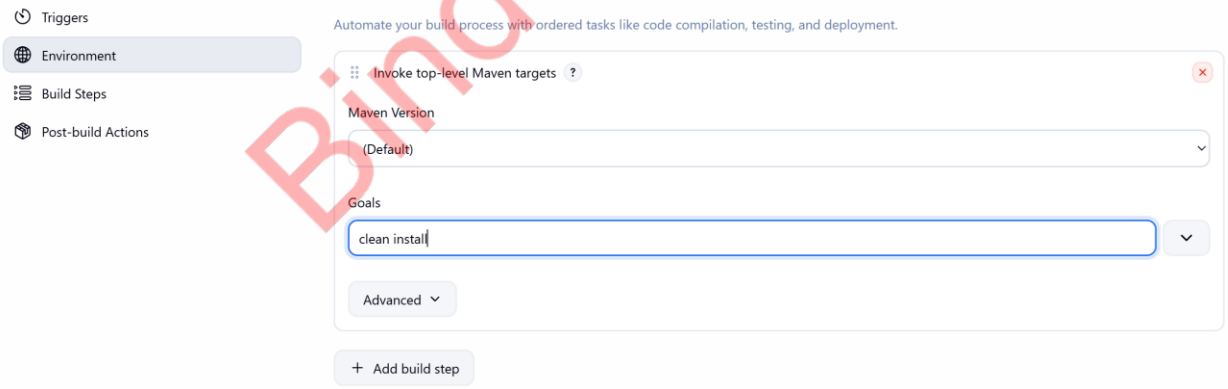



10. Configure Jenkins Build: Scroll to **Build Section, For Maven select Invoke top-level Maven targets**

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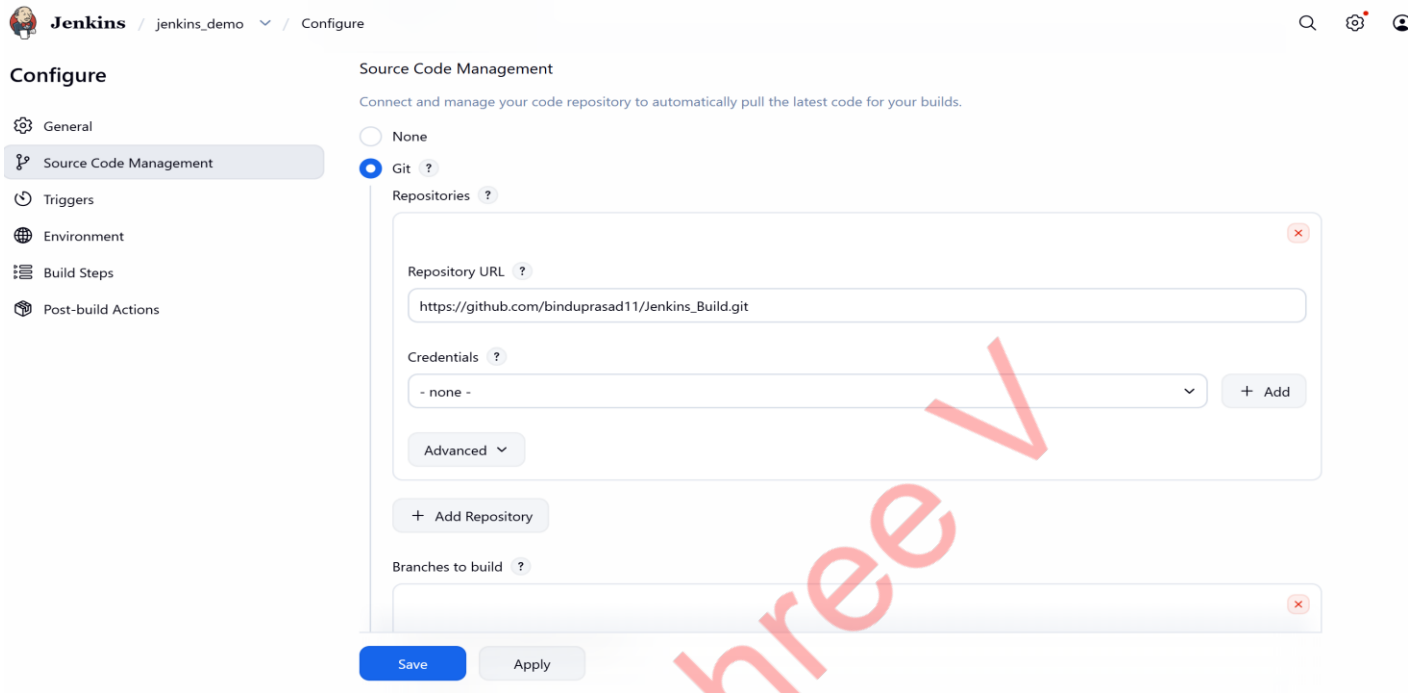



Select Goals clean install




11. Connect Jenkins with IntelliJ Project: Under **Source Code Management, Use your Git Repository**

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ClickBuildNow

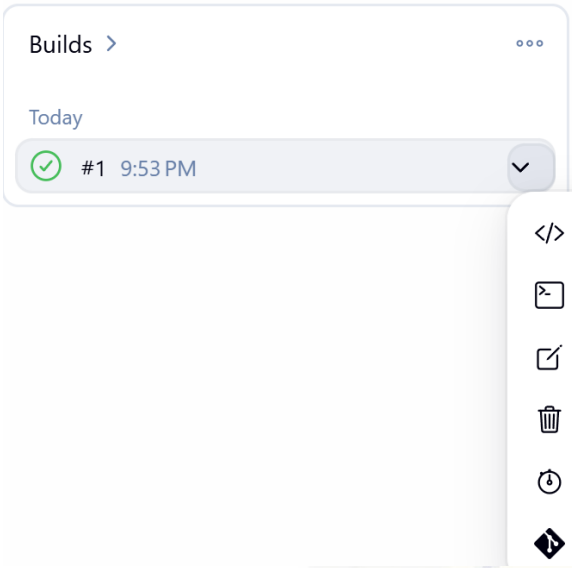



Jenkins will

1. Fetch project
2. Compile code
3. Run tests
4. Generate build

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View Build Results




```
[INFO] Running HelloTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.119 s -- in HelloTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.4.1:jar (default-jar) @ JenkindDemo ---
[INFO] Building jar: C:\Users\BINDUSHREE V\.jenkins\workspace\jenkins_demo\target\JenkindDemo-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- install:3.1.2:install (default-install) @ JenkindDemo ---
[INFO] Installing C:\Users\BINDUSHREE V\.jenkins\workspace\jenkins_demo\pom.xml to C:\Users\BINDUSHREE
V\.m2\repository\com\example\JenkindDemo\1.0-SNAPSHOT\JenkindDemo-1.0-SNAPSHOT.pom
[INFO] Installing C:\Users\BINDUSHREE V\.jenkins\workspace\jenkins_demo\target\JenkindDemo-1.0-SNAPSHOT.jar to
C:\Users\BINDUSHREE V\.m2\repository\com\example\JenkindDemo\1.0-SNAPSHOT\JenkindDemo-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.395 s
[INFO] Finished at: 2026-03-15T21:53:50+05:30
[INFO] ------------------------------------------------------------------------
Finished: SUCCESS
```

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